

Commerce XI Mathematics & Statistics - Part 2

Unit 1: Partition Values

A quick reference based on Unit 1, textbook pages 1-23.

How to use this guide

Use	What to check
Before solving	Identify whether the data are raw, discrete, grouped, less-than cumulative or more-than cumulative.
During solving	Arrange raw data, calculate N and cumulative frequencies, and identify the correct partition class.
Before answering	Check the requested quartile, decile or percentile and include units where required.

Scope

Quartiles - deciles - percentiles - raw and grouped data - interpolation - cumulative frequency - ogives

Textbook Answer Notes

Exercise 1.1, Q9: the verified limits of the central 50% are $(Q_1, Q_3) = (270, 630)$. The textbook answer key gives $(270, 360)$; its Q_3 value is incorrect.

1. Meaning and notation

Measure	Division	Notation	Key interpretation
Quartiles	4 equal parts	Q1, Q2, Q3	25%, 50%, 75% lie below Q1, Q2, Q3
Deciles	10 equal parts	D1 to D9	10i% lie below Di
Percentiles	100 equal parts	P1 to P99	i% lie below Pi
Median	2 equal parts	Q2 = D5 = P50	50% lie below the median

2. Raw and discrete data positions

Partition value	Position after arranging observations in ascending order
Qi	$i(n + 1)/4$
Di	$i(n + 1)/10$
Pi	$i(n + 1)/100$
Fractional position	Interpolate between the two adjacent observations as demonstrated in the textbook

3. Grouped-data formulae

Value	Formula
Quartile	$Q_i = L + (h/f)[iN/4 - \text{c.f.}]$
Decile	$D_i = L + (h/f)[iN/10 - \text{c.f.}]$
Percentile	$P_i = L + (h/f)[iN/100 - \text{c.f.}]$

Here L is the lower boundary of the partition class, h is its class width, f is its frequency, c.f. is the cumulative frequency before it, and N is the total frequency.

4. Choosing the partition class

Required value	Locate this item in cumulative frequency
Q_i	$iN/4$
D_i	$iN/10$
P_i	$iN/100$

5. Ogives

Graph	Plot
Less than ogive	Upper class boundaries against less-than cumulative frequencies
More than ogive	Lower class boundaries against more-than cumulative frequencies
Median	Read at $N/2$, or from the intersection of the two ogives
Quartile	Read Q_1 at $N/4$ and Q_3 at $3N/4$
Decile	Read D_i at $iN/10$
Percentile	Read P_i at $iN/100$

6. Central portions

Required central part	Limits
Central 50%	Q_1 and Q_3
Middle 60%	P_{20} and P_{80}
Middle 40%	P_{30} and P_{70}

7. Final checks

Check	Action
Raw values	Sort before locating a position
Grouped values	Use class boundaries and the preceding cumulative frequency
More-than table	Convert carefully when a less-than cumulative frequency is required
Graphical answer	Treat the reading as approximate
Units	Retain kg, cm, rupees, marks or other stated units